

Network Projects: Packet Tracer

Small Business - Coffee Shop

What devices does it need to get started?

- Workstation PCs and Laptops
- Printers
- Router
- Wireless Access Point (WAP) with Power over Ethernet (PoE) for the main store area. PoE allows the use of a single Ethernet cable for both power and internet.
- Managed Switch (for security) to create VLANs.
- Internet Connection (ISP)

Skills Practiced:

Network planning and topology design
VLAN configuration
Router-on-a-Stick Inter-VLAN Routing
Trunk and Access Port Configuration
DHCP Configuration
SSH Remote Management
PortFast Configuration
Guest Wi-Fi Configuration
Standard Access Control Lists (ACLs)
End-to-End Network Verification

Note: Always remember to save the configuration using the “copy running-config startup-config” command!

Configuring the Switch and Router (Password Encryption)

1. Set the hostname.
2. Disable DNS lookup using the “no ip domain-lookup command”
3. Add security to the Switch and Router: Configure a secret password using “enable secret desert123” for the switch and “enable secret desert1234” for the router in global configuration mode.
4. Encrypt plain-text passwords on the switch using the “service password-encryption” command.
5. Configure a banner using banner “motd # Your Message #”.
6. Secure the console line using line console 0, in global configuration mode (config t) then set a password. Use the “logging synchronous” command in line config mode to prevent the switch from interrupting your typing with system messages.

VLAN Configuration

I created four separate VLANs:

- VLAN 10: Management
- VLAN 20: Sales
- VLAN 30: Guest Wi-Fi
- VLAN 99: Network Configuring (Native/Management)

Creating a VLAN and Assigning Ports

The standard process involves entering global configuration mode, naming the VLAN, selecting the interface, setting it to access mode, and assigning it to the VLAN names I created.

A shorter way to create a new VLAN and assign a range of ports:

```
SW(config)# vlan 10
SW(config-vlan)# name Management
SW(config-vlan)# int range f0/1-5
SW(config-if-range)# sw mod acc
SW(config-if-range)# sw acc vlan 10
```

Trunk Port vs. Multiple Ethernet Cables

I chose to use a single Ethernet cable configured as a trunk port to allow all VLANs to share the same physical link.

What I learned: It's obvious, but I overlooked port speeds when selecting a router. In this setup, the FastEthernet switch ports bottleneck the speed from the ISP router. The switch caps traffic at 100 Mbps. Therefore, buying a 500 Mbps internet plan would be throwing away money because the physical hardware cannot use all of the 500 Mbps.

This bottleneck also impacts the trunk port because all VLANs share the bandwidth of that single port (if using a 1 Gbps port). If multiple devices draw 100 Mbps simultaneously, the bandwidth will quickly add up to the port limit and cause network lag. I am worried about the WAP setup. If the coffee shop gets busy and more than 10 guests connect their laptops while the business VLANs are operational, the network will become sluggish. I need to focus on future proofing network growth to prevent these issues. To improve this I would have to use a GigabitEthernet Port switch.

Trunk Port Commands: Allows multiple Vlan to travel across one physical cable using vlan tags.

```
SW(config)# interface GigabitEthernet 0/1
SW(config-if)# description Trunk to Router ISP
SW(config-if)# switchport trunk encapsulation dot1q
SW(config-if)# switchport mode trunk
SW(config-if)# switchport trunk allowed vlan 10,20,30,99
SW(config-if)# no shut
```

VLAN 99: SSH Remote Management Configuration

VLAN 99 is dedicated exclusively to managing the switch; only network engineers have access. I assigned it the IP address 192.168.99.2 with a default gateway of 192.168.99.1.

```
DesertCoffee-SW(config)# interface vlan 99
DesertCoffee-SW(config-if)# description Switch_management
DesertCoffee-SW(config-if)# ip address 192.168.99.2 255.255.255.0
DesertCoffee-SW(config-if)# no shut
DesertCoffee-SW(config-if)# exit
DesertCoffee-SW(config)# ip default-gateway 192.168.99.1
DesertCoffee-SW(config)# ip domain-name desertcoffee.local
DesertCoffee-SW(config)# username admin privilege 15 secret desert123
DesertCoffee-SW(config)# crypto key generate rsa
DesertCoffee-SW(config)# ip ssh version 2
DesertCoffee-SW(config)# line vty 0 15
DesertCoffee-SW(config-line)# login local
DesertCoffee-SW(config-line)# transport input ssh
DesertCoffee-SW(config-line)# exec-timeout 10
DesertCoffee-SW(config-line)# exit
DesertCoffee-SW(config)# end
```

Verification Commands:

- show vlan brief – Checks active VLANs.
- show ip interface brief – Verifies assigned IP addresses.
- show running-config – Displays the active configuration on the switch or router.

Connecting the Router to the ISP and Switch

Router to ISP:

```
DesertCoffee_Rtr(config)# interface g0/0
DesertCoffee_Rtr(config-if)# description Interface_to_ISP
DesertCoffee_Rtr(config-if)# no shut
```

Router to Switch:

```
DesertCoffee_Rtr(config)# interface g0/1
DesertCoffee_Rtr(config-if)# description Router_to_Switch
DesertCoffee_Rtr(config-if)# no shut
```

Router-on-a-Stick (Inter-VLAN Routing)

When using VLANs, you need Inter-VLAN routing so the different networks can communicate. I am using a single physical cable from the switch to the router, making a Router-on-a-Stick configuration must be done. The only time this wouldn't be required is if I used a Layer 3 switch for routing. This configuration must be repeated for every VLAN sub-interface:

```
DesertCoffee_Rtr(config)# interface g0/1.10
DesertCoffee_Rtr(config-subif)# description Management_office
DesertCoffee_Rtr(config-subif)# encapsulation dot1Q 10
DesertCoffee_Rtr(config-subif)# ip address 192.168.10.1 255.255.255.0
DesertCoffee_Rtr(config-subif)# exit
```

Configure DHCP and Static Addresses

DHCP exclusion

```
DesertCoffee_Rtr(config)# ip dhcp excluded-address 192.168.10.1 192.168.10.20
DesertCoffee_Rtr(config)# ip dhcp excluded-address 192.168.20.1 192.168.20.20
DesertCoffee_Rtr(config)# ip dhcp excluded-address 192.168.30.1 192.168.30.20
```

DHCP Pool

```
DesertCoffee_Rtr(config)# ip dhcp pool Management_Office
DesertCoffee_Rtr(dhcp-config)# network 192.168.10.0 255.255.255.0
DesertCoffee_Rtr(dhcp-config)# default-router 192.168.10.1
DesertCoffee_Rtr(dhcp-config)# dns-server 8.8.8.8
DesertCoffee_Rtr(dhcp-config)# exit
```

(Repeat this process for all pools with their respective IP scopes).

(I also confirmed that when I used a new device in the network, DHCP was working accordingly and gave out appropriate IP addresses according to the VLAN the device was in)

Access Control Lists (ACLs)

Extended ACLs keep unwanted traffic out of secure VLANs. This example restricts VLAN 30 (Guest Wi-Fi) from accessing internal networks while allowing DHCP and general internet access:

```
DesertCoffee_Rtr(config)# ip access-list extended Guest_restrictions
DesertCoffee_Rtr(config-ext-nacl)# permit udp any eq bootpc any eq bootps
DesertCoffee_Rtr(config-ext-nacl)# deny ip 192.168.30.0 0.0.0.255 192.168.10.0 0.0.0.255
DesertCoffee_Rtr(config-ext-nacl)# deny ip 192.168.30.0 0.0.0.255 192.168.20.0 0.0.0.255
DesertCoffee_Rtr(config-ext-nacl)# deny ip 192.168.30.0 0.0.0.255 192.168.99.0 0.0.0.255
DesertCoffee_Rtr(config-ext-nacl)# permit ip 192.168.30.0 0.0.0.255 any
```

Troubleshooting and Lessons Learned

- **VLAN Creation Error:** When trying to create VLANs, the CLI returned an "invalid input detected" error. I realized I was accidentally typing switch commands on the router. Since no configurations were saved yet, I safely wiped the device using write erase and reloaded it to return to factory settings.

- **Physical Layer Scaling:** I made the mistake of not leaving extra physical ports open in each department for future growth. While it isn't an issue right now, I need to remember to plan for spare capacity in future network designs.

- **PortFast and Trunking:** I enabled PortFast on an access port for the guest Wi-Fi. If I plan to use that same physical access point for staff Wi-Fi in the future, the port must be converted to a trunk port so it can tag both the staff and guest VLANs. This saves money on extra hardware.

• **ACL:** Initially, my guest laptops could still ping devices on other secure VLANs. Because I am using a Router-on-a-Stick architecture, applying the ACL to a local switch VLAN interface (interface vlan 30) did not work. The routing happens at the sub-interfaces of the router. To fix it, I had to apply the ACL directly to the router's sub-interfaces. (0/1.10, 0.10.20, 0/1.30, 0/1.99)

The incorrect command I was using was:

```
DesertCoffee_Rtr(config)# interface vlan xx
DesertCoffee_Rtr(config)# ip access-group xxx_restrictions in
```

The command I used to fix that problem:

```
DesertCoffee_Rtr(config)# interface GigabitEthernet0/1.xx
DesertCoffee_Rtr(config-subif)# ip access-group Guest_restrictions in
```

• **Next Steps:** Isolating the networks completely caused my Management PC to lose ping access to the Management VLAN (VLAN 99). While this is standard practice to maximize security, I want to practice tweaking my ACLs to permit one specific IP address from a device to access the network devices via SSH. This will help me practice writing specific host entries within an ACL.

***Understanding the concepts while troubleshooting:**

Troubleshooting why my PC in vlan 10 was not pinging vlan99 192.168.99.2.

***Layer 1** - cables on my pc and switches, routers were all where they are supposed to be and everything is powered on.

***Layer 2** - all my Vlans were configured correctly and running

`show ip interface brief` - confirmed vlan 99 was up and had an IP 192.168.99.2

`show running-config` - confirmed vlan 99 had a default gateway, I also checked the trunk port allowed vlan 99 and vlan 10.

Layer 3 - On the router I ran the "`show running-config`". This showed me the sub interface GigabitEthernet0/1.10 had an ACL applied in the Management_restrictions that explicitly dropped all the traffic from 192.168.10.0/24 network to the 192.168.99.0/24. However, I had an ACL permission in NetworkM_restrictions allowing inbound pings from 192.168.10.21. I forgot to configure it on the Management_restrictions ACL group.

Below was the ACL rule blocking the ping:

```
"deny ip 192.168.10.0 0.0.0.255 192.168.99.0 0.0.0.255"
```

*To change that, the command was used below:

```
DesertCoffee_Rtr# configure terminal
DesertCoffee_Rtr(config)# ip access-list extended Management_restrictions
DesertCoffee_Rtr(config-ext-nacl)# permit ip host 192.168.10.21 192.168.99.0 0.0.0.255
DesertCoffee_Rtr(config-ext-nacl)# end
DesertCoffee_Rtr# write memory
```

***ACL continued and static routing:**

I knew I had to change the PC pinging VLAN 99 to have a different IP. This was because the static IP I used initially was outside the scope of my DHCP exclusion rule. Previously, I used 192.168.10.21, so far it has not cause me any issues, but I did further research and it could cause me problems where the router wants to use the .21 address resulting in collisions; it would be best practice to use an IP within the DHCP exclusion rule I already placed. I changed the IP address to 192.168.10.19 and as expected, my PC was not pinging specifically 192.168.99.2.

I learned that when changing ACL's you have to change the permission on both ends because the traffic will not be allowed inbound or outbound on the desired network.

The commands below were used to change the ACL permissions:

```
DesertCoffee_Rtr# conf t
DesertCoffee_Rtr(config)# ip access-list extended Management_restrictions
```

```
DesertCoffee_Rtr(config-ext-nacl)# no 30
DesertCoffee_Rtr(config-ext-nacl)# 30 permit ip host 192.168.10.19 host 192.168.99.2
DesertCoffee_Rtr(config-ext-nacl)# end
```

```
DesertCoffee_Rtr# conf t
DesertCoffee_Rtr(config)# ip access-list extended NetworkM_restrictions
DesertCoffee_Rtr(config-ext-nacl)# no 20
DesertCoffee_Rtr(config-ext-nacl)# 20 permit ip 192.168.99.2 host 192.168.10.19
DesertCoffee_Rtr(config-ext-nacl)# end
DesertCoffee_Rtr# write
```